Long-Duration Industrial Energy Storage

IonVe ELECTROCHEMICAL REACTORS

IonVe develops Redox Flow Battery systems for industrial applications requiring 4 –12 hours of daily storage. The system architecture has been redesigned from the cell level to achieve a cost and efficiency profile that current products cannot reach.

- MK1 prototype in testing
- MK2 prototype fabrication
- PoC target: June 2026
- Patent applications in preparation

297 €/kWh	−40%	3.4×	600 mA/cm²	> 30 yr
Target installed cost – 6h duration	vs. current VRFB systems	Fewer stacks units at 500 kW output	Target power density	Operating life, no degradation

CONTEXT

01 Long-duration storage is economically viable for industry today

Industrial sites with high demand charges and time-of-use electricity contracts already generate a positive business case for 4–12 hour storage. The barrier is not economics — it is the absence of an affordable, reliable product.

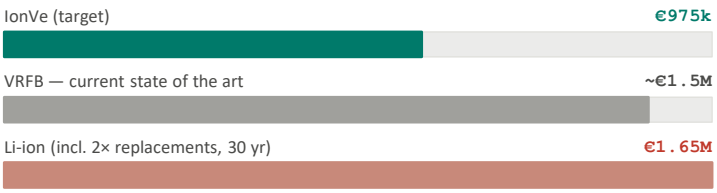
02 Lithium-ion is not cost-effective beyond 4–6 hours

At durations of 6 hours or more, the total cost of ownership of Li-ion over a 20–30 year industrial lifetime is significantly higher than VRFB, due to battery replacements every 10–12 years.

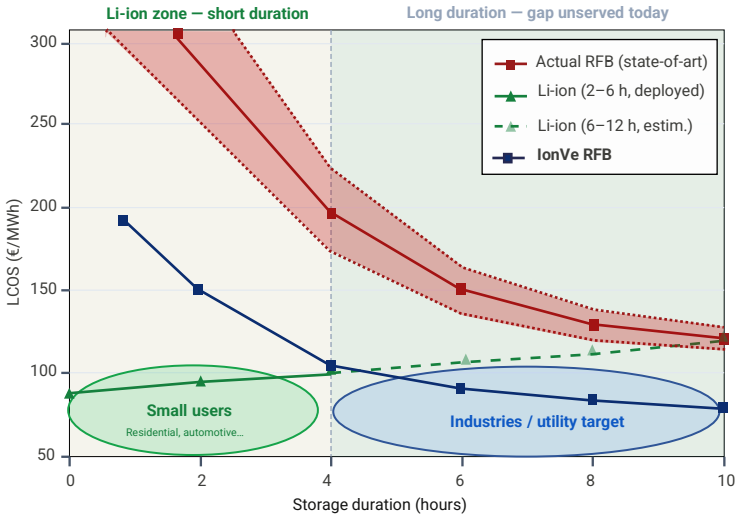
03 Existing RFB systems remain too expensive

Current products are priced at €450–600/kWh installed — a result of an outdated stack and system architecture: high component count, machined graphite parts, and custom engineering for every installation. No standardized catalogue product currently exists.

30-YEAR TCO COMPARISON — 500 kW / 3 MWh



Model output at design-target performance. IonVe recommends Li-ion for durations below 4 hours.



Levelized Cost of Storage of actual storage technology. IonVe outstand for duration > 4 hours.

ARCHITECTURE

INNOVATION 01

Conductive Polymer Frame

A new bipolar plate, based to a new material and architecture aimed at increase efficiency, facilitate assembly, reducing part count and opening a more scalable manufacturing.

Estimated cost reduction: 30–40% per cell vs. machined graphite components

INNOVATION 02

High-Power Ministack Module

A standardized ministack unit achieves +20% power output per module, with 7% higher efficiency. At 500 kW, IonVe system uses 63 versus 212 modules of a conventional RFB, improving manufacturing and simplifying field maintenance.

Target: 600 mA/cm² — ASR:0,15 Ω m⁻²

INNOVATION 03

Modular Architecture

A modular architecture allows higher level of scalability, flow distribution optimization, electrical reconfigurability and hot swap, simplifying power electronics and improving round-trip efficiency.

Estimated efficiency improvement: +5 percentage points RTE

INNOVATION 04

Integrated System Platform

The four innovations are combined into a single co-designed product platform — cell, stack, hydraulics, power electronics, and control — enabling a repeatable, series-producible system.

Power range: 100 kW to 50 MW — same architecture, no re-engineering

+10%	−40%	+20%
Round-trip efficiency	Unit capital cost	Operation range

RISK-GATED PROGRAMME — FOUR BINARY GO/NO-GO GATES

Proof of Concept — Jan–Jun 2026 (↑ We are here)

TRL 3→5

- ✓ MK1/2: Hydraulic- electric tests
- Patent filings (3 applications)
- PoC: 3 go/no-go criteria validated
- ○ Company incorporation

Prototype — Jul–Dec 2026

TRL 5→6

- ○ 2–3 functional prototypes
- ○ 250–500 cycle endurance testing
- ○ Independent external lab validation
- ○ First pilot partner signed

Industrial Pilot — 2027

TRL 6→7

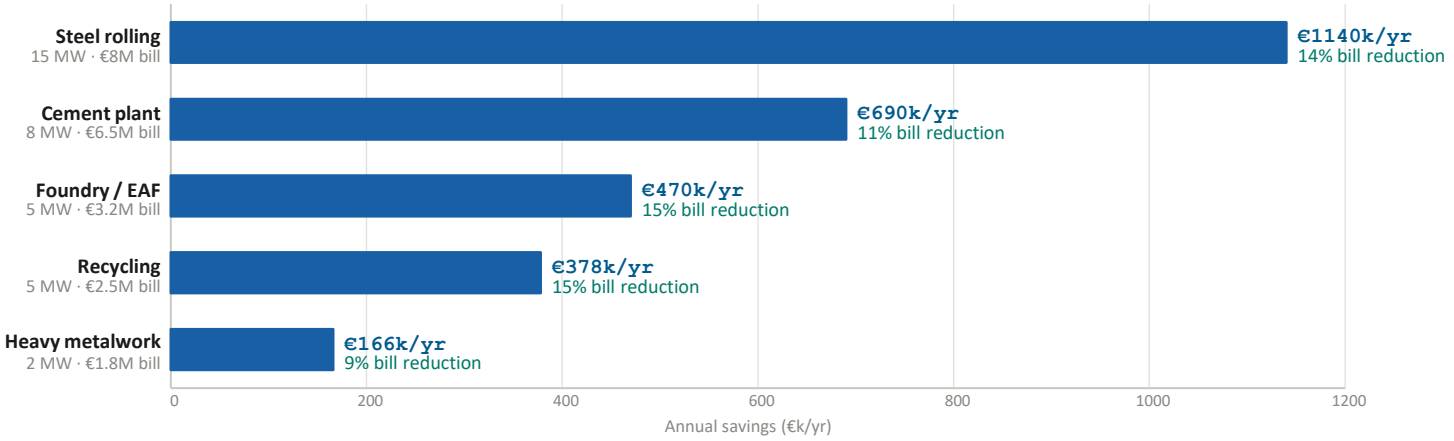
- ○ 2–3 co-development pilots
- ○ Independent performance data
- ○ Supply-chain qualification
- ○ Series A fundraise

Commercial Scale — 2028–2030

TRL 7→9

- ○ 15–30 systems/yr, 5 countries
- ○ EPC channel partnerships
- ○ LTSA recurring revenue
- ○ Energy-as-a-Service model

MODELLED ANNUAL SAVINGS BY INDUSTRIAL SEGMENT



Savings figures are model outputs at design-target system performance. Approximately 65% of savings derive from contractual demand-charge reduction; 35% from time-of-use arbitrage.

REFERENCE CASE — FOUNDRY, 500 kW / 3 MWh

~€470k/yr	€975k	€1.65M	~5 years
Modelled annual savings	30-year TCO — IonVe system	30-year TCO — Li-ion (2× replacements)	Simple payback at target pricing

IonVe is not positioned for every storage application. Lithium-ion remains appropriate for durations below 4 hours and investment horizons shorter than 8 years. IonVe is designed for sites with frequent cycling, high demand charges, and planning horizons of 20–30 years.

REFERENCE SYSTEM CONFIGURATIONS — DESIGN-STAGE SCREENING TOOL

The three configurations below are used as reference cases in IonVe's techno-economic screening model. They represent the design targets and operating points for which the architecture is being optimised, and form the basis of the savings modelling presented in this document.

ENTRY
IonVe 100
100 kW · 600 kWh (6 h standard)

Mid-size industrial sites, pilot installations

STANDARD
IonVe 500
500 kW · 3 MWh (6 h standard)

Foundries, die-casting, cold chain — primary commercial target

INDUSTRIAL
IonVe 1000
1 MW · 6 MWh (6 h standard)

Steel, heavy metalworking, cement, recycling

LTSA: €15,000–30,000/yr per system. Remote monitoring: €3,000–6,000/yr. IonVe can act as electrolyte procurement agent.

IonVe 500 – (500kW – 3MWh / 6h standard)

HOW WE WORK WITH PARTNERS

IonVe is at development stage. No systems are currently available for sale.

01 Site-Specific Screening
No cost · No obligation · NDA

We model annual savings from your 15-min interval load data and electricity contract terms. Output: site-specific report NPV, payback, sensitivity within 2 weeks of data.

→ Site screening report

02 Access to Test Data
On request · Under NDA

We share electrochemical test results and stack performance data as milestones are completed: MK1 hydraulics (done), characterisation (in progress), MK2 results.

→ Technical data package

03 Co-Development Pilot
Post-PoC · Q3 2026 onwards

2–3 pilot installations at cost-plus pricing in exchange for real operating data. Independently verified performance, cost accounting, right of first negotiation.

→ Pilot term sheet — Q1 2027

04 EPC & Integrator Briefings
Open · No prerequisites

Structured technology briefing for project developers and EPC contractors evaluating VRFB. Open to licensing and component supply discussions with parties at scale.

→ 60–90 min, in person or remote

CONTACTS & INFORMATIONS

Davide Bordignon · davide.brdgn@gmail.com · +39 345 172 4876